CprE/SE 4910 WEEKLY REPORT 08

11/04/2026 – 11/10/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

Work was done on the Lightning Talk Presentation and the Design Document v2. Also discussed best architecture approaches for the real application to help with giving the team a more agile workflow in the future.

Past Week Accomplishments

- Group Work:
  - Working on the lighting talk as a group and making sure the whole team was on the same page for testing.
  - Work was done on the design document which included several visuals.
- Nakshatra Gupta:
  - Built a converter script that turns the computer vision team's YOLO label text files into JSON. The script labels each detection by class, marks thin boxes as walls with a start and end point, and can convert a whole folder at once. Tested it on edge cases such as empty files, boxes slightly past the image edge and polygon lines. Anything skipped or adjusted is reported in a warnings list. I worked on lightning talk as well.
- Rishiraj Parmar:
  - Formatted design doc, worked on lighting talk slides, outlined test cases for checking wall dimensions and door placement in the generated 3D model.

- Ryan McNab:
    - Started up on the real project instead of the prototype.
    - Completely rearchitectured the codebase so the app, graphics engine, documentation, and the models live in their own domain.
    - Formatted the pipeline within the application itself in a modular way. Made a directory for each step, including preprocessing, detection, postprocessing, building the json, editing, and rendering with a graphics engine. Each of these steps has a number of substeps which are also placed in the codebase in order.
    - Added interfaces for each major step listed above which the overall pipeline uses to keep things modular. Each step also has a mini-pipeline which uses the substeps within each major step.
    - Implemented the first iteration of the models used in the detection step. One creates box elements created by Daniel and the other was made to complement it. The secondary one finds faces, meaning rooms, in the blueprint and stores them as lists of vectors pointing to each other. Not only were these trained and tested, but implemented too. The faces and box models each have a confidence for each element, and by choosing the most confident from both, it becomes better.
    - Created implementations for each of the interfaces for the major steps and minor steps to fulfill each of the contracts. Each of the steps in the editing pipeline has a full CRUD operation list, and each of the others have a main `run()` method which kicks off the step.
    - Created UI and improved UX for the main application to implement the code from each module. First, there is a file picker and floor list with a button to begin detection. There is then a loading screen, which was taken from the prototype because it was appealing in the feedback. Next, the model returns the JSON and the UI provides a clean recreation for the user to view and correct if mistakes were made. After fixing the walls, windows, doors, stairs, and floor alignment, the JSON is then saved in the local repository and the placeholder graphics engine, which was taken from the prototype, renders it. This placeholder graphics engine will be replaced by Jace in the future, when it gets completed. Among all of these steps, there is a download and import button for the JSON, so the user can share their models with others. There is also a visual pipeline at the top of the screen so the user can know how much longer until the model is viewable. The user can go back steps at any time to edit and tweak any mistakes.
    - Finally, each of the modules of the pipeline were given unit tests to ensure working features. In the lightning talk given by the team, the unit tests were mentioned as a tool to ensure the app works even after replacing an entire module or a substep. On a similar note, multiple evaluation tests were run on the models being used to accurately score each of the models.
    - Created stacked PRs in the team's GitLab repository with all of the above work and explained to the team where everything is, what things do, and how to go forward from here. For more implementation details, view the team repository.
- Daniel Hargrave:

    - Continued training the model with more epochs. Got to a point where the model performance plateaued.
    - Worked on trying to implement a segmentation YOLO model.
        - This led to several issues. It performed very poorly on most of the information.
    - Tried to combine the original model with the segmentation model by ignoring the segmentation model in all cases where there wasn't a diagonal
        - This approach was okay, but it really didn't improve much of the accuracy of the original model
    - Found that the model would very likely not be used in the way that it was originally planned to be.
        - One upside with the segmentation model was that it was really good at distinguishing the different types of doors, which was something that the original model didn't do. This wasn't planned to be something that would be very important until later, but this is still useful.
    - Started working on a model that is more similar to the original model, but includes a theta.
        - It has not finished training yet since it takes a long time. Will test once it is done.
    - Discussed with the team about using a pipeline that uses multiple models. Likely 3 but maybe more.
    - Discussed more post-processing steps that will be needed.
    - Worked on lightning talk.
- Jase Beaubien:
    - Coordinated high-level plans.
    - Helped theorize computer vision solutions.
    - Researched Three.js and began implementing it.
    - Worked on lightning talk.

Pending issues

- Nakshatra Gupta:
    - Class names and image dimensions from the computer vision model are still needed to convert detections into real-world meters in the schema. The schema was built from a single blueprint, so it likely needs changes for features in the other plans, such as multi-plane roofs, porches and stepped floor levels. The move to a multi-model pipeline and a model with a theta value may change which fields can be detected.
- Rishiraj Parmar: N/A
- Ryan McNab:
    - The blueprints that were given to the team differ largely and there are many instances when a window is cut off by a symbol that does not occur on any other windows. There are also many different types of doors that the model needs to be able to track, but that is not very important for the MVP of the project.

    - Half-height walls are not being detected, and instead being seen as full walls. Diagonal walls are also not being detected because the box model is only width and height instead of also with an angle. Daniel will look into that.
    - Stairs cannot be read as diagonal or any specific shape, and they have no changeable height. This means that a little step up by a foot will cause a full size stair to be generated. There are also U-shape stairs which need to be stuck at different heights and connect together, but the model reads it as a huge rectangle surrounding the width and height fully.
- Daniel Hargrave:
    - Models are taking a really long time to train. Need to try and find a way to speed up this process, since currently breaks are required to take continuously throughout the process.
    - Still haven't been able to test the newest model, since it hasn't finished training.
- Jase Beaubien:
    - N/A

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Built a YOLO text to JSON converter and tested it on the computer vision team's label file and worked on lightning talk.	7	31
Rishiraj Parmar	Outlined test cases for checking wall dimensions and door placement in the generated 3D model.	5	30
Ryan McNab	Created the entire pipeline of the project with a waterfall approach to allow for easy iterative agile iterations in the future. Trained a secondary model used for postprocessing steps. Implemented all of the UI needed for the MVP of the project.	15	48
Daniel Hargrave	Worked on the models significantly. Finished	8	38

	training the original model, trained and tested a segmentation model, tried to combine them together, and finally started work on a model more similar to the original that uses a theta value.		
Jase Beaubien	Pivoted to Three.js and did research on the library, worked on rendering in it, and helped communicate ideas for computer vision pipeline.	7	32

Comments and Extended Discussion

Plans for the upcoming week

- Nakshatra Gupta:
    - Revise the schema to cover features from the other sample plans. Get class names and image size from the computer vision team and map detections into the schema format.
- Rishiraj Parmar:
    - Identify how to check for gaps between connected walls, overlapping geometry, and incorrect wall alignment in the generated 3D model.
- Ryan McNab:
    - Implement a gallery similar to the prototype and implement a graceful failure.
- Daniel Hargrave:
    - Will finish training/testing/likely modifying model that uses theta.
    - Will work more on post-processing steps that were unable to be worked on over the last week.
- Jase Beaubien:
    - Implement interface between the computer vision pipeline to the graphics engine.